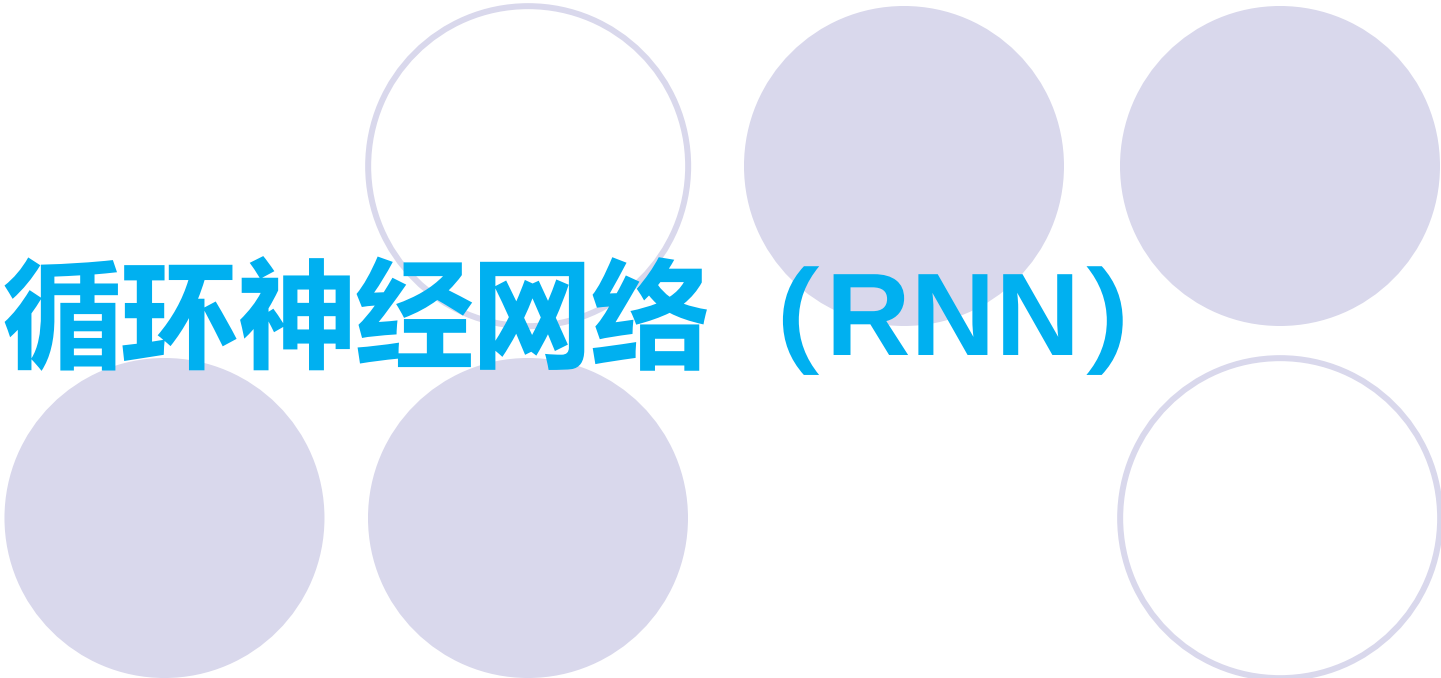




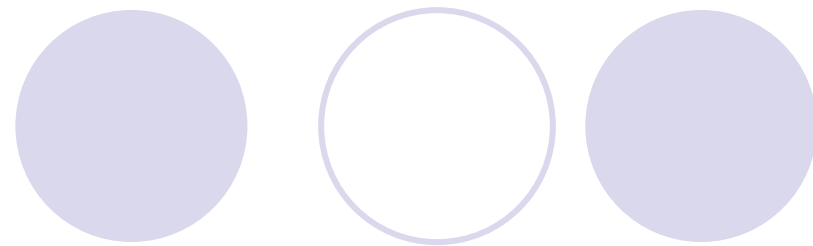
循环神经网络 (RNN)

主要内容

The slide features decorative circles in the top header area. On the left, there is a solid light purple circle and an outlined light purple circle. On the right, there are three circles: a solid light purple circle, an outlined light purple circle, and another solid light purple circle.

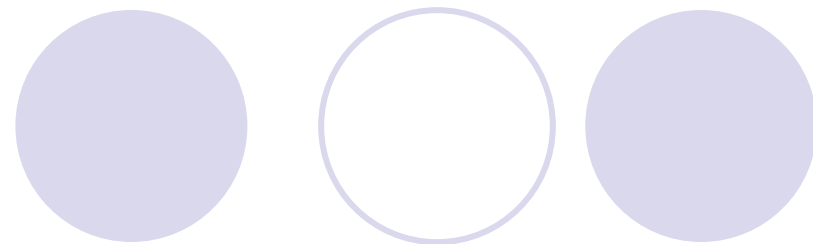
- 简单循环神经网络
- LSTM（长短时记忆网络）
- Transformer
- 应用简介

循环神经网络



- 对序列进行建模（对比HMM）
- 循环网络在每一个时序位置上的具有相同的结构（“循环”，对比GMM-》HMM）
- 应用领域
 - NLP
 - 分类、序列标注、语言模型
 - 语音处理
 - 语音识别、去噪
 - 时间序列预测
 - 天气预报、金融分析

简单循环神经网络



- 问题提出

- 时间序列数据的基本问题：给定时序向量序列 $x_1, x_2, \dots, x_T$
对它们的边缘概率 $p_1, p_2, \dots, p_T$

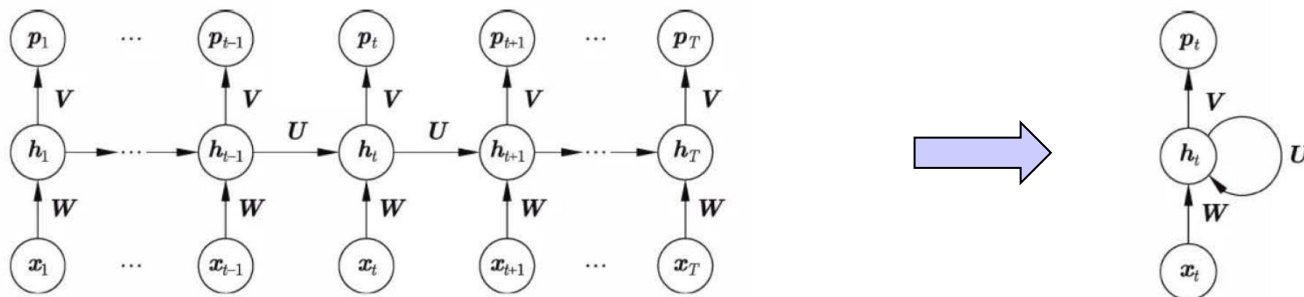
- 为什么不用前馈网络

- 输入的序列长度（时间轴）不固定
- 即便长度固定，普通前馈网络对相邻重复模式出现在不同位置需要分辨表示和学习
 - 造成模式种类多
 - 参数空间大
 - 没有充分利用条件独立性（对比HMM）

- 解决思路

- 每个时序位置建立相同的网络结构（减少参数）
- 引入隐状态、并将相邻的神经网络连接（引入条件独立）

简单循环神经网络



$$h_t = \tanh(U \cdot h_{t-1} + W \cdot x_t + b) \quad p_t = \text{softmax}(V \cdot h_t + c)$$

p_t 可以描述一个离散变量的概率分布
上述公式还可以描述如下

$$r_t = U \cdot h_{t-1} + W \cdot x_t + b$$

$$h_t = \tanh(r_t)$$

$$z_t = V \cdot h_t + c$$

$$p_t = \text{softmax}(z_t)$$

简单循环神经网络特点

- 当前位置输出依赖
 - 当前输入（短距离依赖）
 - 前一位置隐状态（实质是以前所有输入信息，长距离依赖）
- 是一种自回归模型（auto-regressive model）
- 是更一般的动态系统（dynamical system）的一种实现方式

$$s(t) = F(x(t), s(t-1))$$

$$y(t) = G(s(t))$$

- x 为输入， s 为状态， y 为输出

简单循环神经网络的学习

- 估计模型中的参数

- 输入: $x_1, x_2, \dots, x_T$

- 输出: $y_1, y_2, \dots, y_T$

$$P(y_1, y_2, \dots, y_T | x_1, x_2, \dots, x_T) = \prod_{t=1}^T P(y_t | x_1, x_2, \dots, x_t)$$

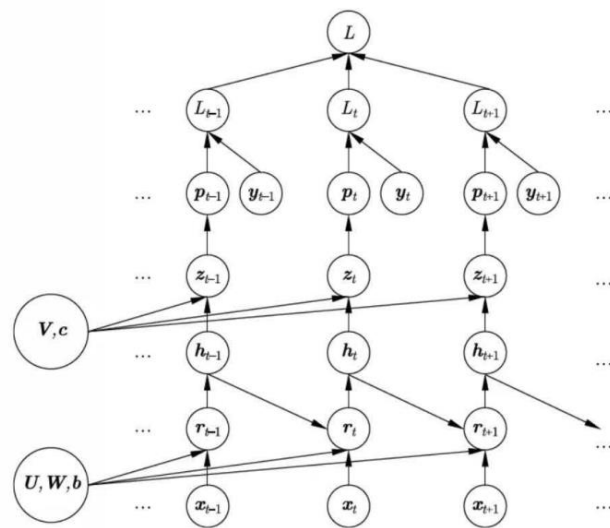
$$L = \sum_{t=1}^T L_t = - \sum_{t=1}^T \log P(y_t | x_1, x_2, \dots, x_t)$$

$$\theta \leftarrow \theta - \eta \cdot \frac{\partial L}{\partial \theta}$$

- 关键是计算 $\frac{\partial L}{\partial \theta}$

- 简单循环神经网络的计算图

- 关键是计算 $\frac{\partial L}{\partial z_t}$ $\frac{\partial L}{\partial r_t}$



简单循环神经网络的学习

$$\frac{\partial L}{\partial z_t} = \frac{\partial L}{\partial L_t} \frac{\partial L_t}{\partial z_t} = y_t - p_t, \quad t = 1, 2, \dots, T$$

$$\frac{\partial L}{\partial r_T} = \frac{\partial z_T}{\partial r_T} \frac{\partial L}{\partial z_T} = \frac{\partial h_T}{\partial r_T} \frac{\partial z_T}{\partial h_T} \frac{\partial L}{\partial z_T} = \text{diag}(1 - \tanh^2 r_T) \cdot V^T \cdot \frac{\partial L}{\partial z_T}$$

$$\begin{aligned} \frac{\partial L}{\partial r_t} &= \frac{\partial r_{t+1}}{\partial r_t} \frac{\partial L}{\partial r_{t+1}} + \frac{\partial z_t}{\partial r_t} \frac{\partial L}{\partial z_t} = \frac{\partial h_t}{\partial r_t} \frac{\partial r_{t+1}}{\partial h_t} \frac{\partial L}{\partial r_{t+1}} + \frac{\partial h_t}{\partial r_t} \frac{\partial z_t}{\partial h_t} \frac{\partial L}{\partial z_t} \\ &= \text{diag}(1 - \tanh^2 r_t) \cdot U^T \cdot \frac{\partial L}{\partial r_{t+1}} + \text{diag}(1 - \tanh^2 r_t) \cdot V^T \cdot \frac{\partial L}{\partial z_t}, \\ &\quad t = T-1, \dots, 2, 1 \end{aligned}$$

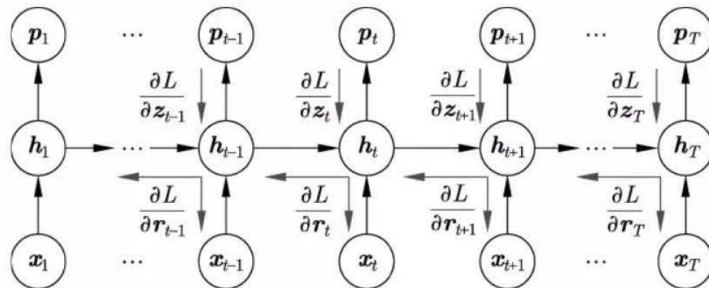
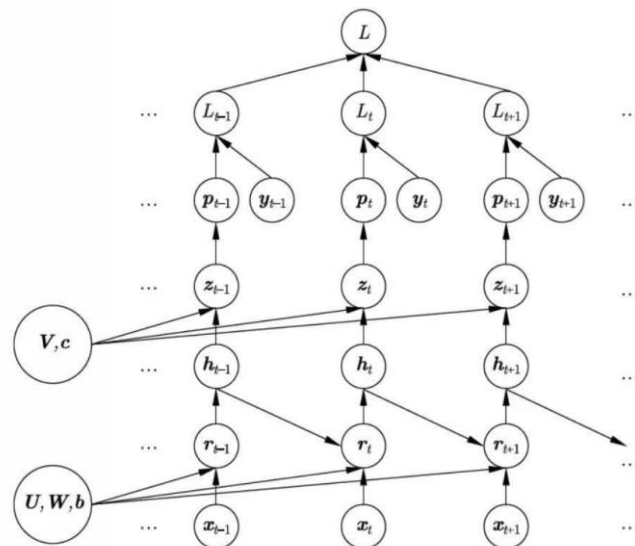
$$\frac{\partial L}{\partial c} = \sum_{t=1}^T \frac{\partial z_t}{\partial c} \frac{\partial L}{\partial z_t} = \sum_{t=1}^T \frac{\partial L}{\partial z_t}$$

$$\frac{\partial L}{\partial V} = \sum_{t=1}^T \frac{\partial z_t}{\partial V} \frac{\partial L}{\partial z_t} = \sum_{t=1}^T \frac{\partial L}{\partial z_t} \cdot h_t^T$$

$$\frac{\partial L}{\partial b} = \sum_{t=1}^T \frac{\partial r_t}{\partial b} \frac{\partial L}{\partial r_t} = \sum_{t=1}^T \frac{\partial L}{\partial r_t}$$

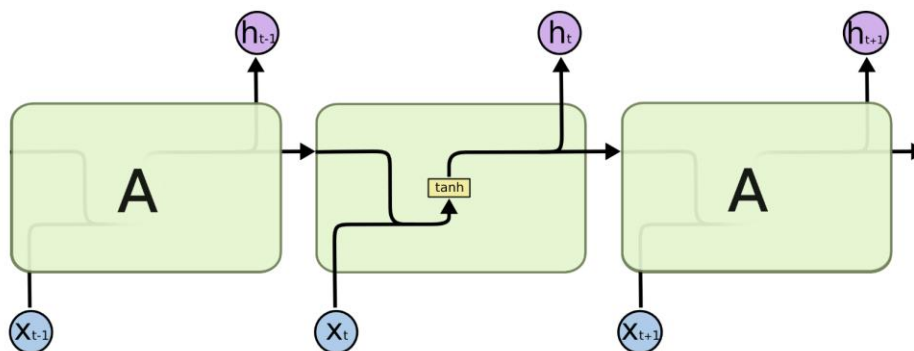
$$\frac{\partial L}{\partial U} = \sum_{t=1}^T \frac{\partial r_t}{\partial U} \frac{\partial L}{\partial r_t} = \sum_{t=1}^T \frac{\partial L}{\partial r_t} \cdot h_{t-1}^T$$

$$\frac{\partial L}{\partial W} = \sum_{t=1}^T \frac{\partial r_t}{\partial W} \frac{\partial L}{\partial r_t} = \sum_{t=1}^T \frac{\partial L}{\partial r_t} \cdot x_t^T$$

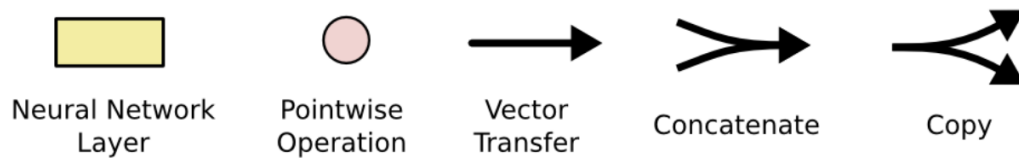
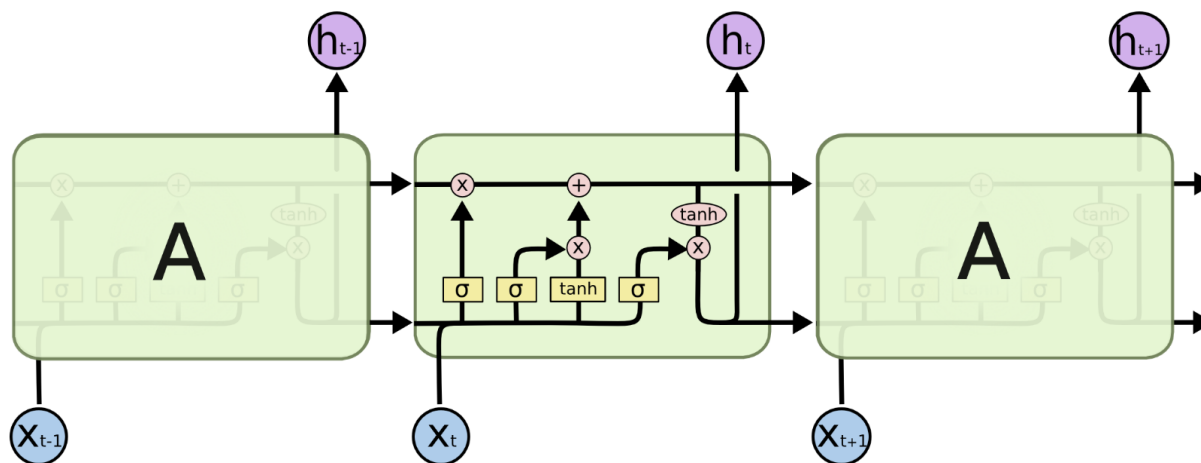


长短时记忆网络

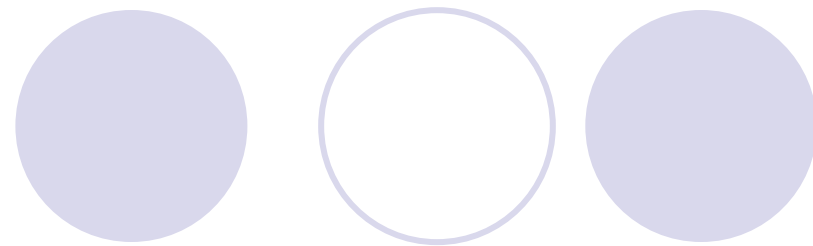
- 简单循环神经网络面临的问题
 - 梯度爆炸（对梯度进行尺度变换，成比例缩小）
 - 对于一些近距离依赖，简单RNN问题不大，如
the clouds are in the sky
 - 但对于长距离依赖，会碰到一些问题，如
I grew up in France...I speak fluent French
- 实践证明，简单RNN对长距离效果一般
- LSTM
 - 显式地避免长距离依赖



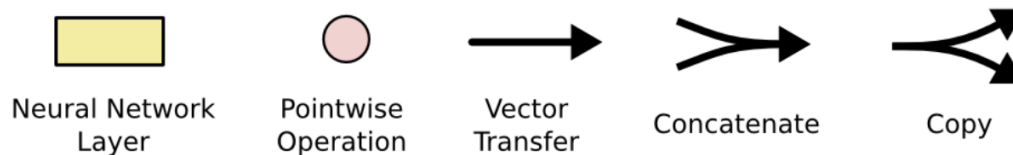
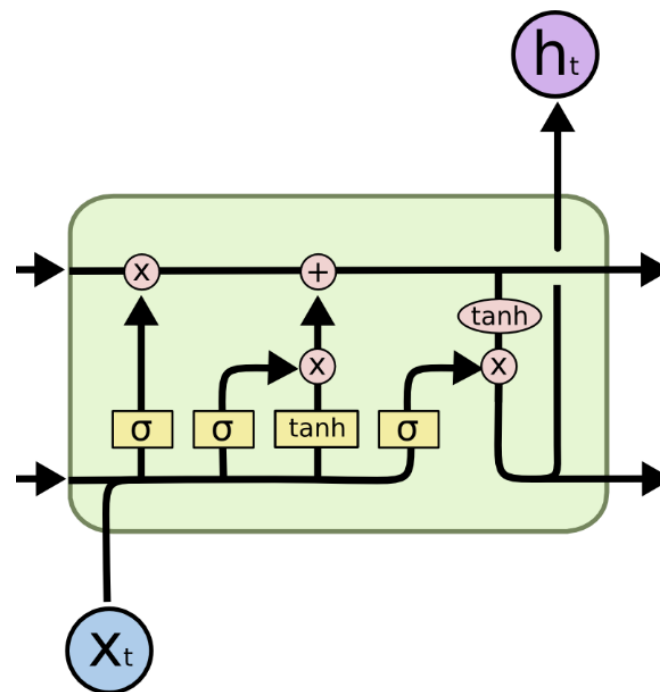
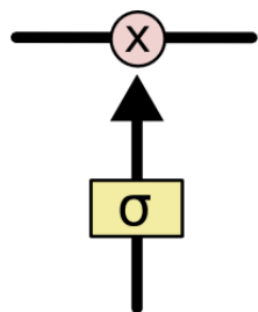
长短时记忆网络



长短时记忆网络

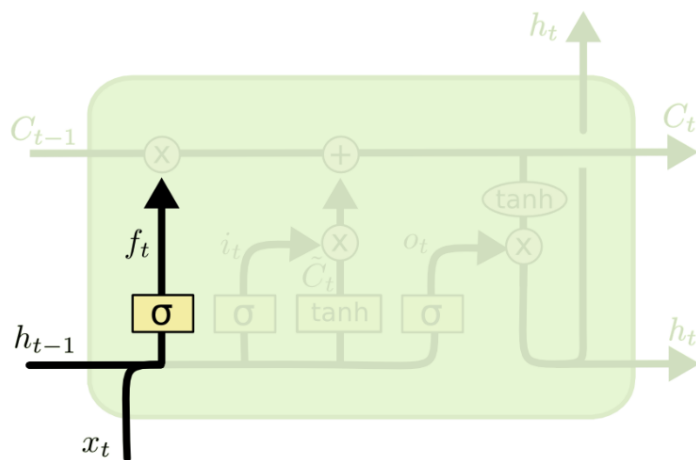


- 核心思想：引入记忆单元和门控
- 门控是一种输出在0-1之间

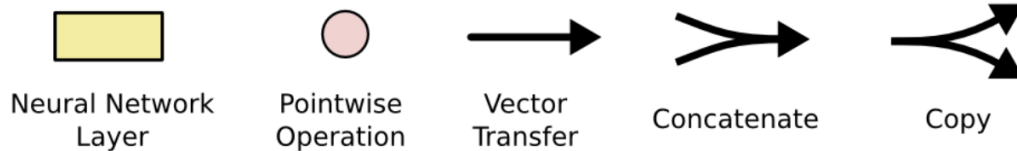


遗忘门

- 确定从记忆元中删除哪些信息

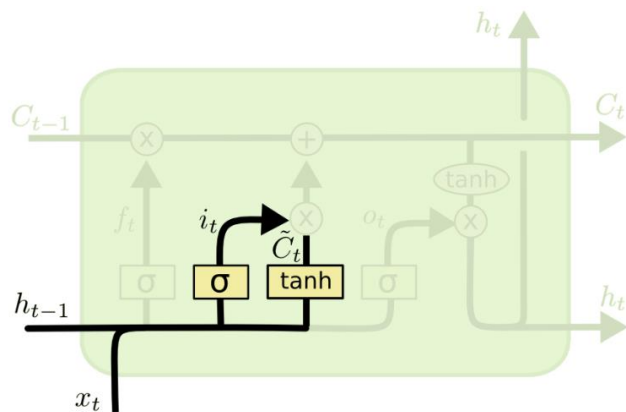


$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$

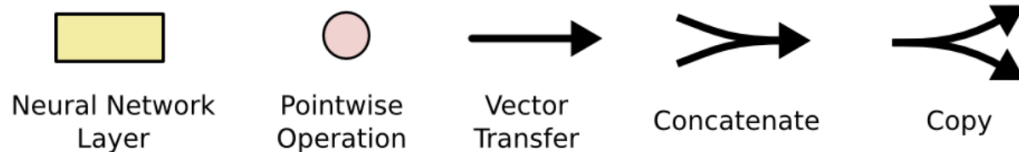


输入门

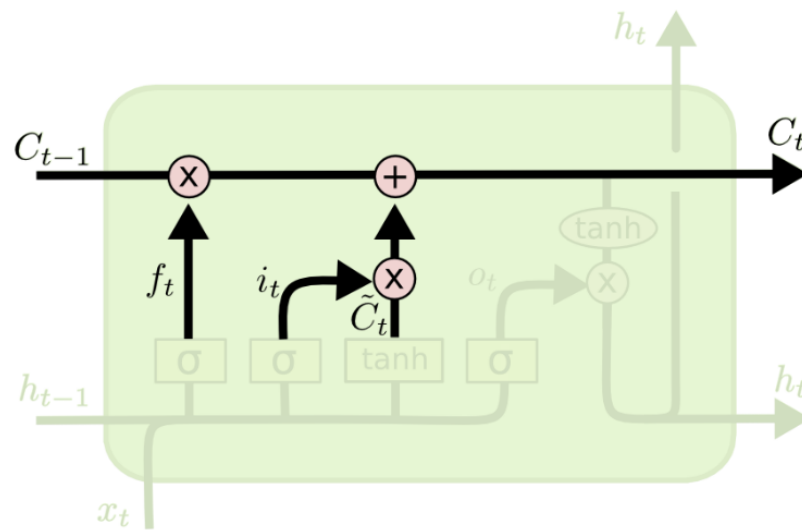
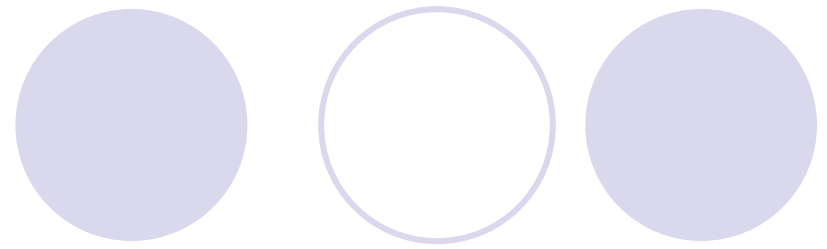
- 确定把哪些信息存入记忆元中



$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$
$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C)$$



更新记忆元



$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t$$

Neural Network
Layer

Pointwise
Operation

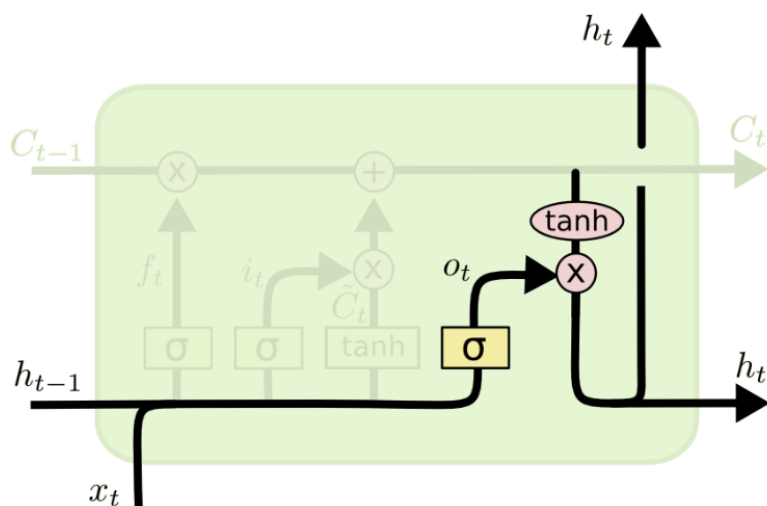
Vector
Transfer

Concatenate

Copy

输出门

● 确定隐层

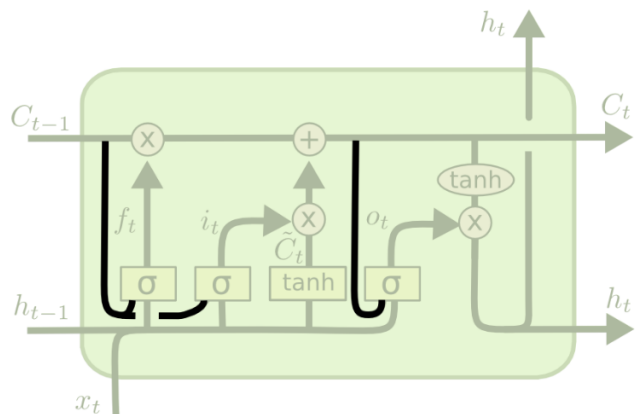


$$o_t = \sigma(W_o [h_{t-1}, x_t] + b_o)$$

$$h_t = o_t * \tanh(C_t)$$

LSTM的一些变形

- 确定门状态时，需考虑记忆元信息



$$f_t = \sigma(W_f \cdot [C_{t-1}, h_{t-1}, x_t] + b_f)$$

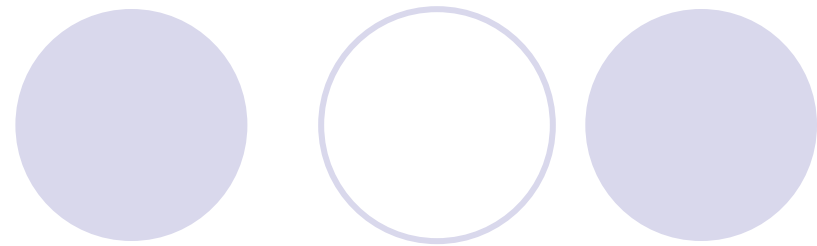
$$i_t = \sigma(W_i \cdot [C_{t-1}, h_{t-1}, x_t] + b_i)$$

$$o_t = \sigma(W_o \cdot [C_t, h_{t-1}, x_t] + b_o)$$

Transformer

- RNN一类网络问题
 - 按照位置串行计算，效率低（虽然有一些分解trick）
 - 对长距离依赖建模还是或多或少地存在问题
- 注意力机制
 - 直接建立各位间的依赖关系，不考虑它们的距离
 - 避免按位置依次计算，为并行创造条件
- Transformer
 - 第一个只采用Attention的序列转换模型
 - 避免需要对其（aligned）循环结构
 - 无需卷积操作

Transformer的结构



- 目的
 - 从一个序列变为另一个序列（如语言翻译）

- 输入

$$(x_1, \dots, x_n)$$

- 输出

$$(y_1, \dots, y_m)$$

- 输入的向量表示

$$\mathbf{z} = (z_1, \dots, z_n)$$

Transformer的结构

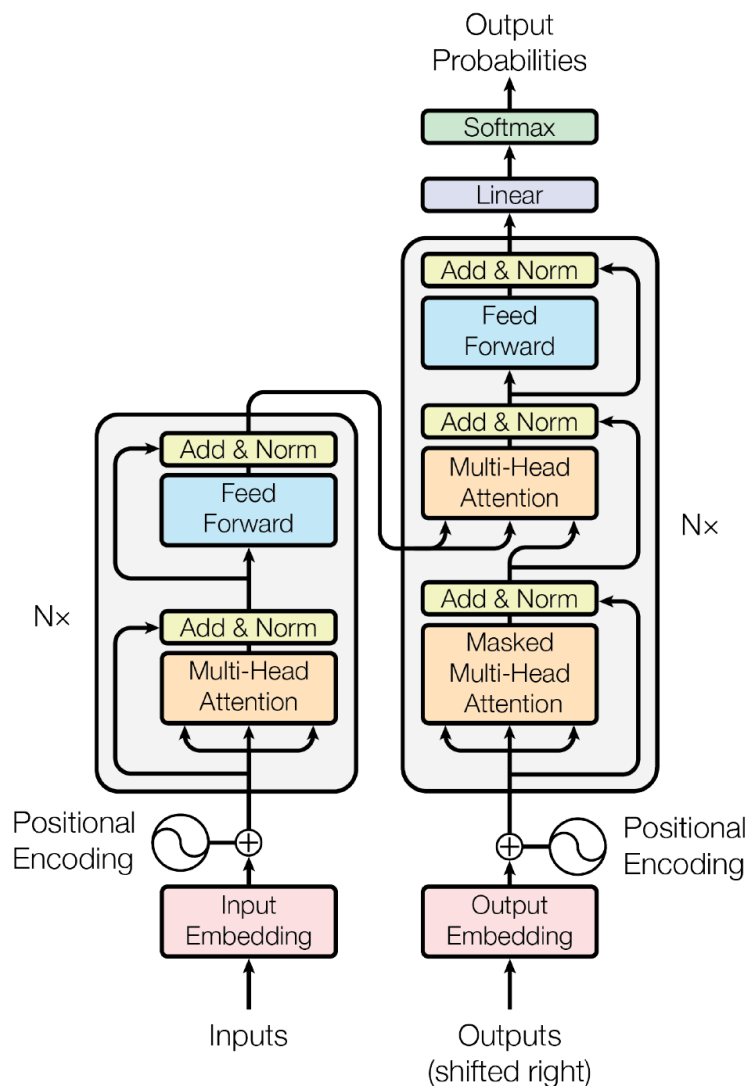
● 编码器

- N=6个相同的层
- 每个层有2个子层，子层间采用残差连接
- 层归一化
- 多头注意力
- 子层和嵌入层的输出都是512维

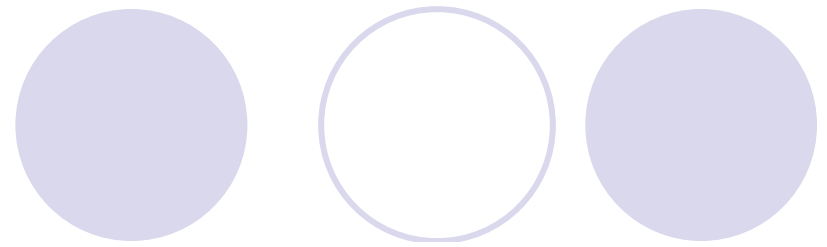
$$\text{LayerNorm}(x + \text{Sublayer}(x))$$

● 解码器

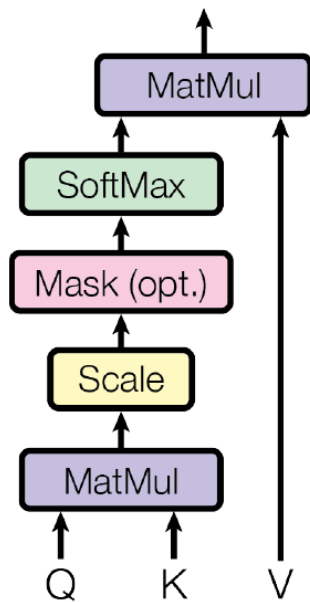
- N=6个相同的层
- 3个子层
- 残差连接
- 多头注意力
- 解码层的注意力要保证输出当前位置结果时只依赖于其前面已生成数据



注意力 (Attention)

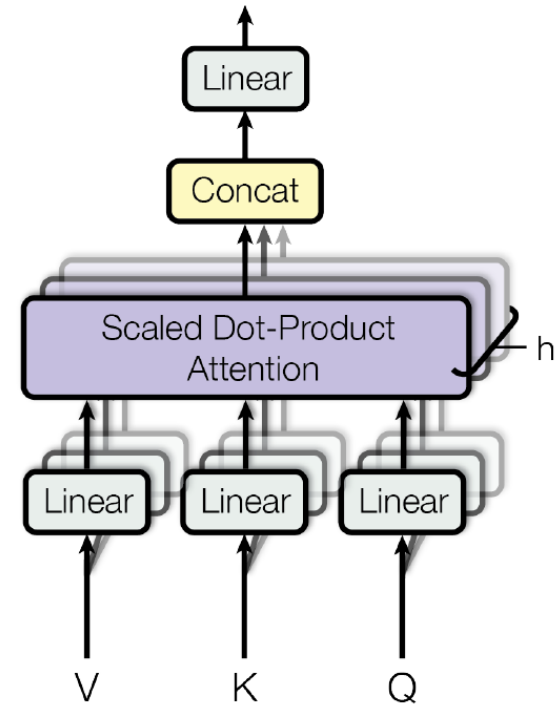


Scaled Dot-Product Attention



$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Multi-Head Attention



$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O$$

where $\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$

● 自注意力的一些优势

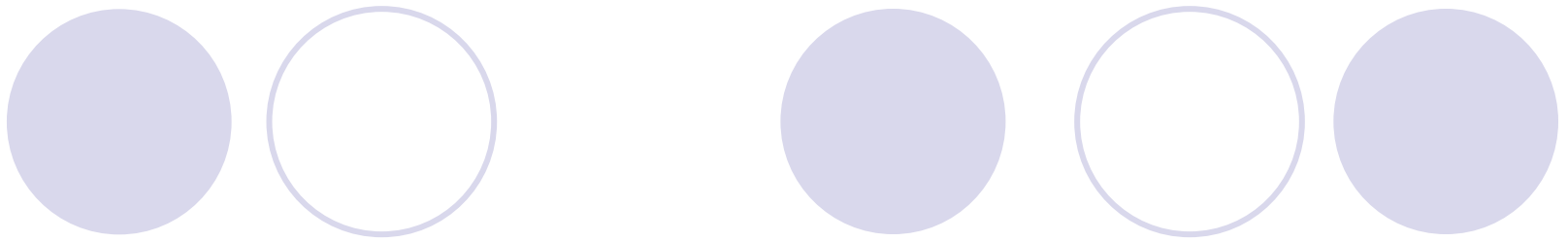
Layer Type	Complexity per Layer	Sequential Operations	Maximum Path Length
Self-Attention	$O(n^2 \cdot d)$	$O(1)$	$O(1)$
Recurrent	$O(n \cdot d^2)$	$O(n)$	$O(n)$
Convolutional	$O(k \cdot n \cdot d^2)$	$O(1)$	$O(\log_k(n))$
Self-Attention (restricted)	$O(r \cdot n \cdot d)$	$O(1)$	$O(n/r)$

● 位置编码

- 因为没有采用循环结构或者卷积

$$PE_{(pos, 2i)} = \sin(pos/10000^{2i/d_{\text{model}}})$$

$$PE_{(pos, 2i+1)} = \cos(pos/10000^{2i/d_{\text{model}}})$$



Attention Visualizations

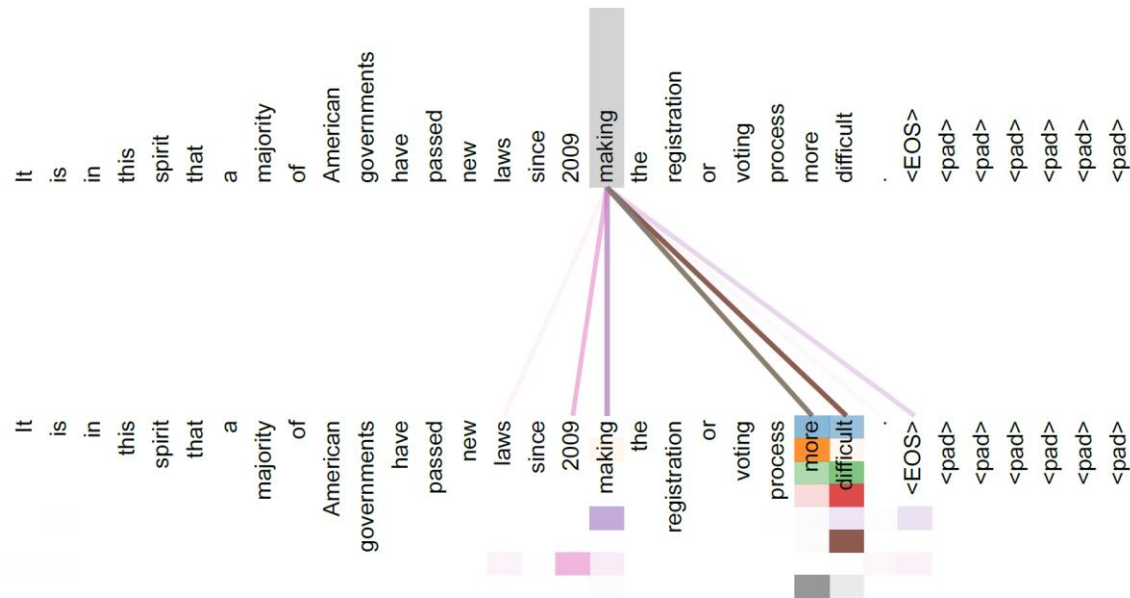


Figure 3: An example of the attention mechanism following long-distance dependencies in the encoder self-attention in layer 5 of 6. Many of the attention heads attend to a distant dependency of the verb 'making', completing the phrase 'making...more difficult'. Attentions here shown only for the word 'making'. Different colors represent different heads. Best viewed in color.

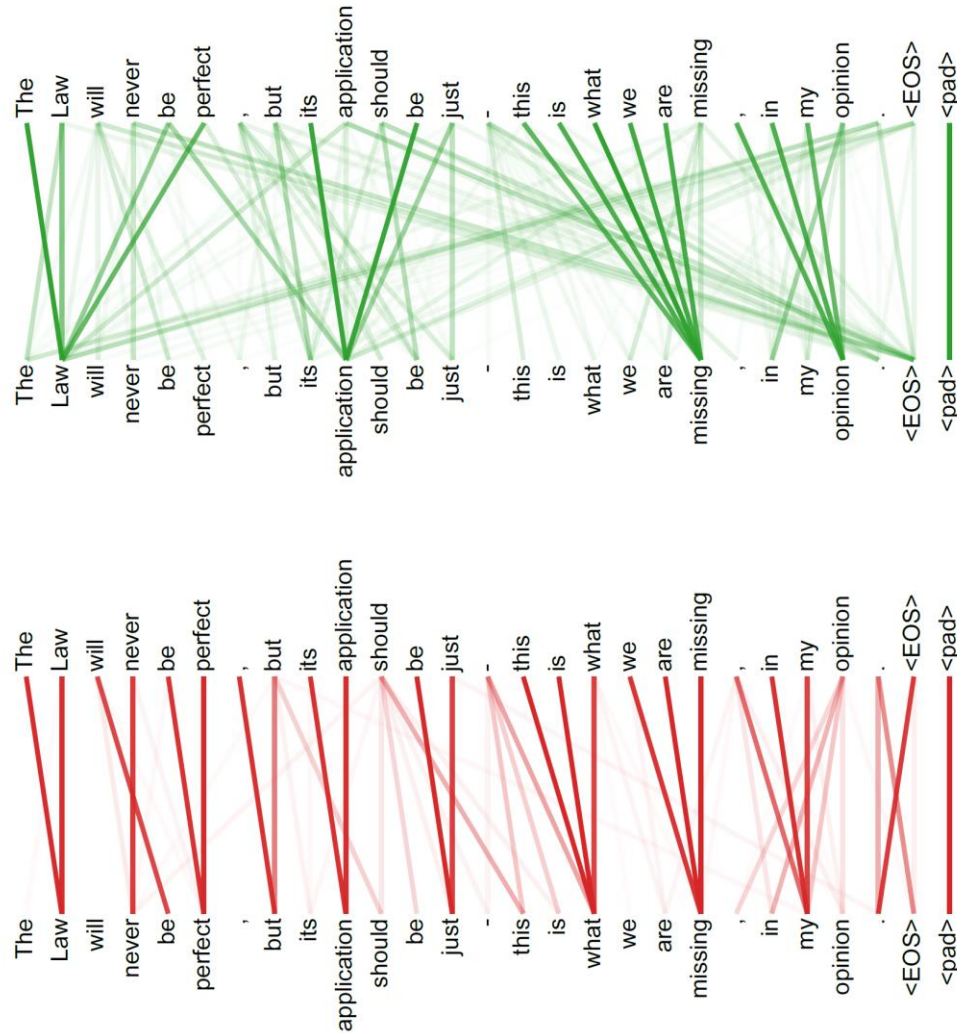


Figure 5: Many of the attention heads exhibit behaviour that seems related to the structure of the sentence. We give two such examples above, from two different heads from the encoder self-attention at layer 5 of 6. The heads clearly learned to perform different tasks.